INSTRUCTOR HANDBOOK • DO NOT DISTRIBUTE TO STUDENTS

RetailPulse — Trainer Assignment Guide & Answer Key Summary

How to Distribute, Roll Out, Support in MySQL Workbench, and Grade Student Submissions

1. How to Assign This Project to Your Students

1. **What to Share with Students:** Zip and share **only** the `Student_Assignment_Kit/` folder containing:
   - `RetailPulse_Student_Assignment.pdf` (the complete question packet)
   - `00_student_schema.sql` and `01_load_csv_data.sql`
   - `csv_data/` (all 9 CSV files: `regions.csv` through `returns.csv`)
2. **What to Keep Private (Trainer Answer Key):** Keep root files `02_seed_data.sql` through `07_sample_insights_report.md` private as your reference solutions!
3. **Suggested 2-Week Rollout Schedule (4 Checkpoints):**

Milestone	Scope	Recommended Classroom Delivery
Session 1 (Day 1–2)	Setup + Part A (A1–A5)	Walk through the 9-table ERD; students import CSVs in parent-to-child order and run the 5 data quality audit checks.
Session 2 (Day 3–6)	Part B (Q1–Q12)	Teach why joining `orders` + `order_items` + `returns` directly inflates totals; students solve Executive KPIs & Campaign ROAS.
Session 3 (Day 7–11)	Part C (Q13–Q24)	Lab on Window Functions (`NTILE`, `LAG`, `DENSE_RANK`, `ROWS BETWEEN`), RFM, Cohort Retention, and Market Basket self-joins.
Session 4 (Day 12–14)	Part D + E + Review	Students wrap logic into Views/Procedures/Trigger, export a Workbench ER diagram PNG, and present 5 executive recommendations.

> **Solution File Mapping:** Tasks A1–A5 → `03_data_cleaning_and_validation.sql` • Q1–Q12 → `04_exploratory_and_kpi_analysis.sql` • Q13–Q24 → `05_advanced_analytics_rfm_cohorts.sql` • B1–B8 → `06_views_procedures_triggers.sql`.

2. Verified Benchmark Answers (MySQL 8.0.46 Reference Outputs)

Use these verified reference numbers to quickly grade student submissions without re-running every query manually:

Question / Task	Expected Verified Output Highlights
Part A (A1–A5)	Row counts: `regions=5, customers=30, categories=6, suppliers=6, products=24, campaigns=8, orders=72, order_items=159, returns=10`. All 5 audit checks must report `0 | PASS`.
Q1 (Exec KPIs)	`72` orders (`63` Delivered, `4` Cancelled = `5.56%`, `3` Returned), `30` active buyers, GMV `$9,121.00`, Refunds `$864.00`, **Realized Net Revenue `$7,584.00`**, COGS `$4,105.00`, **Gross Profit `$3,479.00` (`45.87%` margin)**, AOV `$124.24`.
Q2–Q3 (Trends)	`34` monthly buckets (`2024-01` to `2026-10`). Peak month: `2024-11` (`$903.00` revenue). YoY Q3 growth rebounds to `+54.79%` in `2026-Q3` (`$856.00` vs `$553.00` in `2025-Q3`).
Q4 (Categories)	Top revenue: **Electronics** (`$3,003.00` net, `36.43%` share, `42.72%` margin). Highest margin: **Beauty & Personal Care** (`68.30%`). Only category below target: **Fashion & Apparel** (`42.86%` vs `48.00%` target = `-5.14 pts`).

Question / Task	Expected Verified Output Highlights
Q5 (Regions & SLA)	#1 Region: **South Asia** (`$3,633.00` net, `3.45%` late). Highest SLA problem: **Europe** (`44.44%` late deliveries, avg `6.37` days vs 5-day SLA). **Southeast Asia**: `0.00%` late.
Q6 (Payments)	Top methods: `Credit Card` (`$3,362.00`) & `UPI` (`$2,995.00` , `0%` cancel/return). Highest risk: `COD` (`33.33%` cancellation + `33.33%` return rate).
Q7 (Top Products)	#1 `EL-101 ProSound ANC Headphones` (`$1,341.00` net, `$521.00` profit). #2 `EL-102 ErgoMech Keyboard` (`$990.00`). Margin anomaly: `HK-303 SmartAir HEPA Purifier` ranks **#7** in revenue (`$402.00`) but drops to **#22** in profit (`$62.00` , `15.42%` margin due to returns).
Q8–Q9 (Suppliers & Discounts)	Worst supplier return rate: `LoomCraft Textiles` (`27.78%`). Discounted orders (`42`) have higher AOV (`$134.79` vs `$107.19`) but lower margin (`48.88%` vs `56.55%` full price).
Q10–Q12 (Campaigns, Returns, DOW)	Top campaign ROAS: `Autumn Productivity 2026` (Email, `4.36x` ROAS, `104.64%` ROI, `$1,220.00` rev). Top return drivers: `Late Delivery` (`$328.00`) & `Size / Fit Issue` (`$216.00`); `2` damaged write-offs (`$191.00`). Peak order day: `Sunday` (`12` orders); peak revenue day: `Monday` (`$1,479.00`).
Q13–Q15 (CLV & RFM)	Top 3 LTV: #1 `Aarav Sharma` (`$1,299.00` , 8 orders), #2 `Liam Carter` (`$1,002.00`), #3 `Ananya Iyer` (`$835.00`). RFM Segments across 27 active buyers: **Champions (`6` , `56.60%` rev)**, Loyal (`5`), Hibernating/Lost (`5`), About to Sleep (`4`), At Risk (`3`), Potential Loyalists (`2`), Needs Attention (`2`); plus `3` zero-purchase signups.
Q16–Q18 (Cohorts & Churn)	`2024-Q1` cohort (`6` buyers): `50.0%` Q1, `33.3%` Q2–Q4, `66.7%` Q5+ retention. High Churn Risk (>2x cycle): `Emma Cole` (`4.79x`), `Zara Al-Mansoor` (`3.38x`), `Sophia Bennett` (`2.44x`).
Q19–Q21 (ABC, Basket, Rolling)	`10` Class A SKUs (~67.4% rev), `7` Class B, `7` Class C. Top Basket Pair: `Vitamin C Serum + Ceramide Moisturizer` (`8` co-purchases, `4.18` lift). Highest lift: `Pour-Over Kettle + Glass Storage Set` (`10.20` lift).
Q22–Q24 (Channels, Stock, Funnel)	Best LTV channel: `Referral` (`$545.75` avg LTV, `100%` repeat). **4 SKUs flagged REORDER NOW:** `FA-204` (15/20), `SF-504` (12/20), `EL-104` (18/25), `BP-403` (14/30). Funnel: `30` signups → `27` (≥1 order, `90.0%`) → `18` (≥2, `66.67%`) → `10` (≥3+, `55.56%`).

3. Common Student Pitfalls to Watch For During Grading

- **Join Fan-Out Error:** Joining `orders` → `order_items` → `returns` directly and calling `SUM(o.order_discount)` multiplies the order discount by the number of line items! Require CTE pre-aggregation at `order_id` grain.
- **Forgetting `order_status <> 'Cancelled'` :** Including the 4 cancelled orders inflates revenue and distorts AOV and RFM frequency.
- **`NTILE(5)` Direction Mistake on Recency:** Students often sort `recency_days ASC` , accidentally giving the most recent buyers score `1` instead of `5` . Verify `ORDER BY recency_days DESC` !
- **MySQL 8.0 `ONLY_FULL_GROUP_BY` Errors:** Remind students that every non-aggregated column in `SELECT` must appear in `GROUP BY` (or use `CROSS JOIN` for single-row reference date CTEs).